

The UV–projected row, integrated

$\int d^2r$ and $\int d\xi$ done — the UV term and the finite terms

$$\int d^2r e^{iq \cdot r} \frac{1}{\xi \bar{D}} \frac{s \cdot P}{P^2 r^2} \left[\xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\bar{\xi}} \right], \text{ then } \int d\xi$$

0. The answer

$$\frac{d^3 N_{2\text{NLO}}}{d^3 k} = \frac{\alpha_s^2}{8\pi^5 k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \underbrace{\frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2}}_{\text{WW} \times \text{WW}} \left[\mathcal{U} + \mathcal{F} \right] \left[\dots \right] \rho^{b'}(x') \rho^f(y),$$

$$\mathcal{U} = \left(2\ell_k - \frac{11}{6} \right) \log \frac{(y - x)^2}{m^2}$$

$$\mathcal{F} = \frac{1 + e^{ik \cdot (y-x)}}{2} \ell_k^2 + c_0 \ell_k + \frac{\pi^2}{6} - \frac{67}{36} + c_1$$

with $\ell_k = \log(k^+/\Lambda)$, $\alpha_s = g^2/4\pi$, and c_0, c_1 the two finite functions of k and $y - x$ given in §4. The whole m –dependence — equivalently the whole $1/\epsilon_{\text{UV}}$ — is the single term $\mathcal{U}$, and $-\frac{11}{6} \times 2N_c = -b$ at $N_f = 0$.

1. What has to be integrated

With $r = z - x$, $s = y - x$, $P = x' - w'$, $\bar{\xi} = 1 - \xi$, $q = \bar{\xi}k$,

$$D = (r - s)^2 + M^2, \quad M^2 = \frac{\xi}{\bar{\xi}} s^2, \quad D_0 \equiv D|_{r=0} = s^2 + M^2 = \frac{s^2}{\bar{\xi}}, \quad C_{\text{UV}}(\xi) = \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}}{2N_c}.$$

Nothing in the bracket depends on r except $1/r^2$ and $1/D$, and C_{UV} does not depend on r at all. So the entire transverse integration is *one* master,

$$\mathcal{G}(\xi) \equiv \int d^2r \frac{e^{iq \cdot r}}{(r^2 + m^2) [(r - s)^2 + M^2]},$$

m being the ultraviolet regulator ($z \rightarrow x$); nothing else needs one.

2. The z integral, exactly

Feynman–parametrise, $\frac{1}{AB} = \int_0^1 \frac{d\alpha}{[\alpha A + \bar{\alpha} B]^2}$ with $A = r^2 + m^2$, $B = D$:

$$\alpha A + \bar{\alpha} B = (r - b)^2 + \Delta, \quad b = \bar{\alpha} s, \quad \Delta = \bar{\alpha}(\alpha s^2 + M^2) + \alpha m^2.$$

Shift $r = b + u$; the phase factorises, $e^{iq \cdot r} = e^{i\bar{\alpha} q \cdot s} e^{iq \cdot u}$, and the u integral is the two–dimensional Yukawa transform differentiated once,

$$\int \frac{d^2u e^{iq \cdot u}}{(u^2 + \Delta)^2} = -\frac{\partial}{\partial \Delta} 2\pi K_0(q\sqrt{\Delta}) = \frac{\pi q}{\sqrt{\Delta}} K_1(q\sqrt{\Delta}), \quad q \equiv |q|.$$

Writing $t = \bar{\alpha}$ (so $\Delta \rightarrow m^2$ at $t = 0$, which is where the divergence sits),

$$\mathcal{G} = \pi \int_0^1 dt e^{itq \cdot s} \frac{q K_1(q\sqrt{\Delta})}{\sqrt{\Delta}}, \quad \Delta = t[(1 - t)s^2 + M^2] + (1 - t)m^2.$$

As $t \rightarrow 0$: $\Delta \rightarrow tD_0 + m^2$, $xK_1(x) \rightarrow 1$, $e^{itq \cdot s} \rightarrow 1$, so the integrand tends to $1/(tD_0 + m^2)$. Add and subtract it; $\pi \int_0^1 \frac{dt}{tD_0 + m^2} = \frac{\pi}{D_0} \log \frac{D_0 + m^2}{m^2}$, and the remainder is finite at $m = 0$:

$$\mathcal{G} = \frac{\pi}{D_0} \left[\log \frac{D_0}{m^2} + \hat{I}(\xi) \right] + O(m^2),$$

$$\hat{I}(\xi) = D_0 \int_0^1 dt \left[e^{itq \cdot s} \frac{q K_1(q\sqrt{\Delta_0})}{\sqrt{\Delta_0}} - \frac{1}{t D_0} \right], \quad \Delta_0 = t[(1 - t)s^2 + M^2].$$

The integrand is integrable at $t \rightarrow 0$ (the bracket tends to $i q \cdot s / D_0$) and regular at $t \rightarrow 1$ for $M > 0$. Equivalently, in closed form up to one α -integral,

$$\hat{I} = \log \frac{4e^{-2\gamma_E}}{q^2 D_0} + \frac{2 s \cdot \Psi - \Phi}{\pi}, \quad \Phi = 2\pi e^{iq \cdot s} K_0(Mq), \quad \Psi^i = \int d^2 r \frac{e^{iq \cdot r} r^i}{r^2 D},$$

which agrees with the t -representation to 10^{-14} (§6A) — the two were derived independently.

3. Putting it back in the row

Multiply by $C_{UV}(\xi) \frac{s \cdot P}{P^2}$ and by the $\frac{1}{\xi}$ of the longitudinal denominator. Because $\frac{1}{D_0} = \frac{\bar{\xi}}{s^2}$, the $\bar{\xi}$ cancels exactly:

$$\frac{1}{\bar{\xi}} \frac{s \cdot P}{P^2} C_{UV}(\xi) \mathcal{G} = \pi C_{UV}(\xi) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \left[\underbrace{\log \frac{(y - x)^2}{\bar{\xi} m^2}}_{UV} + \underbrace{\hat{I}(\xi)}_{\text{finite}} \right].$$

So after the z integration the row is a product of two Weizsäcker–Williams kernels times $P_{gg}/2N_c$ times a logarithm, plus the same kernels times $P_{gg}/2N_c$ times the finite function $\hat{I}$. The choice of D_0 as the argument of the logarithm is a scheme choice; it is the one that makes the UV logarithm read $\log [(y - x)^2 / \bar{\xi} m^2]$, i.e. the running coupling at the dipole size. Any other choice moves a constant (and a $\log \bar{\xi}$) between the two brackets.

4. The p^+ integral

$\int \frac{d\xi}{2\pi}$ over $[\lambda, 1 - \lambda]$, $\lambda = \Lambda/k^+$, $\ell_k = \log(1/\lambda)$. The two pieces behave quite differently.

4.1 The UV term — elementary

$\log \frac{(y-x)^2}{\xi m^2} = \log \frac{(y-x)^2}{m^2} + \log \frac{1}{1-\xi}$, and both moments of $C_{UV} = \xi(1-\xi) - 2 + \frac{1}{\xi(1-\xi)}$ are elementary:

piece of C_{UV}	from	$\int d\xi$	$\int d\xi \log \frac{1}{1-\xi}$
$\xi(1-\xi)$	Θ_1 , instantaneous	$\frac{1}{6}$	$\frac{5}{36}$
$\frac{\xi}{1-\xi}$	Θ_2	$\ell_k - 1$	$\frac{1}{2}\ell_k^2 - 1$
$\frac{1-\xi}{\xi}$	Θ_4	$\ell_k - 1$	$\frac{\pi^2}{6} - 1$
total		$2\ell_k - \frac{11}{6}$	$\frac{1}{2}\ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36}$

$$\int_{\lambda}^{1-\lambda} d\xi C_{UV}(\xi) \log \frac{(y-x)^2}{\xi m^2} = \left(2\ell_k - \frac{11}{6}\right) \log \frac{(y-x)^2}{m^2} + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36}.$$

The double logarithm comes only from Θ_2 , the $\pi^2/6$ only from Θ_4 , and it is the instantaneous Θ_1 that turns -2 into $-\frac{11}{6}$ and -2 into $-\frac{67}{36}$. Note that $\frac{67}{18} - \frac{\pi^2}{3} = -2\left(\frac{\pi^2}{6} - \frac{67}{36}\right)$ is the constant of Eq. (C.10) of arXiv:1610.03453.

4.2 The finite term

$\hat{I}$ depends on ξ through $M^2 = \frac{\xi}{\xi} s^2$ and through $q = \bar{\xi} k$, so $\int d\xi C_{UV} \hat{I}$ is not elementary. Its structure is fixed by the two endpoints.

$\xi \rightarrow 0$. Then $M^2 \rightarrow 0$ and $\Delta_0 \rightarrow ts^2(1-t+\xi)$ develops a logarithm at $t \rightarrow 1$: $\hat{I} = e^{ik \cdot s} \log \frac{1}{\xi} + c_0 + O(\xi \log \xi)$, with

$$c_0(k, s) = s^2 \int_0^1 dt \left[e^{itk \cdot s} \frac{k K_1(ks\sqrt{t(1-t)})}{s\sqrt{t(1-t)}} - \frac{1}{ts^2} - \frac{e^{ik \cdot s}}{s^2(1-t)} \right],$$

a convergent integral (the second subtraction handles $t \rightarrow 1$, the first $t \rightarrow 0$). Since $C_{UV} \rightarrow 1/\xi$ there, $\int d\xi C_{UV} \hat{I}$ picks up $e^{ik \cdot s} \frac{1}{2} \ell_k^2 + c_0 \ell_k$.

$\xi \rightarrow 1$. Then $M \rightarrow \infty$ and $q \rightarrow 0$ together, $\hat{I} \sim \bar{\xi} \log \frac{1}{\xi} \rightarrow 0$, so $C_{UV} \hat{I} \sim \log \frac{1}{\xi}$ — integrable. This endpoint contributes no rapidity logarithm.

$$\int_{\lambda}^{1-\lambda} d\xi C_{UV}(\xi) \hat{I}(\xi) = e^{ik \cdot (y-x)} \frac{\ell_k^2}{2} + c_0 \ell_k + c_1,$$

c_1 being the remaining convergent double integral. For the test kinematics $s = (0.7, -0.4)$, $k = (0.9, 0.6)$ the fit over $\lambda = 10^{-3} \dots 10^{-6}$ gives (§6D)

$$\begin{aligned} \text{coefficient of } \frac{1}{2} \ell_k^2 : & 0.925384 + 0.380329i \quad \text{vs} \quad e^{ik \cdot s} = 0.924909 + 0.380188i, \\ c_0 = & -0.588515 - 0.116237i \quad (\text{closed form}), \quad c_1 = 0.350743 + 0.149676i. \end{aligned}$$

5. Assembling

Collecting the prefactor, $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4 k^+} \times \frac{\pi}{2\pi} = \frac{g^4}{128\pi^7 k^+} = \frac{\alpha_s^2}{8\pi^5 k^+}$, and adding §4.1 and §4.2:

$$\begin{aligned} \frac{d^3 N_{2\text{NLO}}}{d^3 k} &= \frac{\alpha_s^2}{8\pi^5 k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} [\mathcal{U} + \mathcal{F}] [\dots] \rho^{b'}(x') \rho^f(y) \\ \mathcal{U} &= \left(2\ell_k - \frac{11}{6}\right) \log \frac{(y-x)^2}{m^2} \longleftrightarrow \left(2\ell_k - \frac{11}{6}\right) \left[\frac{1}{\epsilon_{UV}} + \log(\pi e^{\gamma_E} \mu^2 (y-x)^2) \right], \\ \mathcal{F} &= \underbrace{\frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36}}_{\text{from the UV log's } \bar{\xi}} + \underbrace{e^{ik \cdot (y-x)} \frac{\ell_k^2}{2} + c_0 \ell_k + c_1}_{\text{from } \hat{I}} = \frac{1 + e^{ik \cdot (y-x)}}{2} \ell_k^2 + c_0 \ell_k + \frac{\pi^2}{6} - \frac{67}{36} + c_1. \end{aligned}$$

Three things to read off.

- **The UV term.** Its coefficient is $\int d\xi P_{gg}/2N_c = 2\ell_k - \frac{11}{6}$, and $-\frac{11}{6} \cdot 2N_c = -\frac{11N_c}{3} = -b$ at $N_f = 0$: a product of two WW kernels times $b \log$, i.e. the running coupling at the dipole size $(y-x)^2$. This is the piece that cancels against the $-2/\epsilon$ of the self-energy counterterm.
- **The rapidity logarithms.** ℓ_k^2 appears twice, once from each endpoint: the $\frac{1}{2} \ell_k^2$ of §4.1 comes from $\xi \rightarrow 1$ (through $\log \frac{1}{\xi}$ against the $\frac{\xi}{1-\xi}$ pole), the $e^{ik \cdot (y-x)} \frac{1}{2} \ell_k^2$ from $\xi \rightarrow 0$ (through the $z \rightarrow y$ logarithm $\log \frac{1}{M^2}$ against the $\frac{1-\xi}{\xi}$ pole). They are the $(\delta Y)^2/\text{LO}^2$ content of the row and must be subtracted before the NLO kernel is read off.
- **Scheme.** Only $\mathcal{U} + \mathcal{F}$ is fixed; the boundary between them is the argument chosen for the logarithm in §3. Keeping $2\pi K_0(m|q|)$ whole instead of splitting at D_0 would move $\log \frac{1}{\xi^2}$ across, changing $\frac{1}{2} \ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36}$ into $\ell_k^2 + \frac{\pi^2}{3} - \frac{67}{18}$ and c_0, c_1 correspondingly. The coefficient of $\log(1/m^2)$, $2\ell_k - \frac{11}{6}$, is the same in every scheme.

6. Checks

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kinematics: s = y-x = [ 0.7 -0.4]   k = [0.9 0.6]   P = x'-w' = [1.3 0.5]
            s^2 = 0.6500   k.s = 0.3900   e^(i k.s) = +0.924909+0.380188j   (s.P)/(P^2 s^2) = 0.563045

A. the z integral.  G = int d^2r e^{iq.r}/[(r^2+m^2)((r-s)^2+M^2)] = (pi/D0)[ln(D0/m^2) + Ihat]
(i) representation vs direct 2D quadrature
  xi=0.30 m=1e-03: rep +48.90822797+1.38855418j   quad +48.90821825+1.38854040j   rel 3.4e-07
  xi=0.65 m=1e-03: rep +24.70758925+0.25775835j   quad +24.70759484+0.25775663j   rel 2.4e-07
(ii) Ihat vs the independent route ln(4 e^-2g/(q^2 D0)) + (2 s.Psi - Phi)/pi
  xi=0.65: +0.1712658056+0.1523730591j   +0.1712658056+0.1523730591j   rel 1.1e-14
  xi=0.30: +0.7145718579+0.4104197719j   +0.7145718579+0.4104197719j   rel 2.6e-15
  xi=0.10: +1.6203600571+0.7910772424j   +1.6203600571+0.7910772424j   rel 3.6e-15

B. Ihat(xi), the finite part left by the z integral
  xi=0.9   Ihat = +0.012075+0.039452j
  xi=0.7   Ihat = +0.128399+0.127295j
  xi=0.5   Ihat = +0.340010+0.239992j
  xi=0.3   Ihat = +0.714572+0.410420j
  xi=0.1   Ihat = +1.620360+0.791077j
  xi=0.01  Ihat = +3.682646+1.639524j
xi -> 0 : Ihat = e^(i k.s) ln(1/xi) + c0 + 0(xi ln xi)
  xi=1e-03: Ihat - e^(iks) ln(1/xi) = -0.586932-0.115577j
  xi=1e-04: Ihat - e^(iks) ln(1/xi) = -0.588316-0.116155j
  xi=1e-05: Ihat - e^(iks) ln(1/xi) = -0.588491-0.116227j
xi -> 1 : Ihat ~ xibar ln(1/xibar) -> 0, so C_UV*Ihat stays integrable (no extra l_k)
  xibar=1e-02: Ihat = -3.4897e-03+3.8951e-03j   C_UV*Ihat = -0.345552+0.385690j
  xibar=1e-03: Ihat = -7.8945e-04+3.8986e-04j   C_UV*Ihat = -0.788661+0.389476j
  xibar=1e-04: Ihat = -1.2275e-04+3.8998e-05j   C_UV*Ihat = -1.227353+0.389939j

C. the xi integrals of C_UV = Pgg/(2Nc)   (mpmath, 30 digits, lambda = 1e-9)
  int C_UV      = 39.613198342559   2 l_k - 11/6      = 39.613198340559
  int C_UV ln(1/xib) = 214.51069645084   l_k^2/2 + pi^2/6 - 67/36 = 214.51069643011

D. the xi integral of the finite part, int dxi C_UV(xi) Ihat(xi)
  lambda=1e-03 l_k= 6.9078 : +18.324371+8.409181j
  lambda=1e-04 l_k= 9.2103 : +34.127344+15.195038j
  lambda=1e-05 l_k=11.5129 : +54.838511+23.997916j
  lambda=1e-06 l_k=13.8155 : +80.454071+34.816703j
fit to a l_k^2/2 + b l_k + c :
  a = +0.925384+0.380329j   e^(i k.s) = +0.924909+0.380188j   (a must be e^(i k.s))
  b = -0.594243-0.117932j   c0      = -0.588491-0.116227j   (b must be c0)
  c = +0.350743+0.149676j

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zint/src/uv_master.py — the master $\mathcal{G}$ and $\hat{I}$; zint/src/run_uvint.py — every number above, written to zint/RESULTS_uvint.txt; zint/src/quad2d.py — the direct oscillatory 2D quadrature used to check it.